OFFICIAL WORKSHOP PROGRAM · 30 OCTOBER 2026

DroneCom 2026

Drone-Assisted Wireless Communications for 5G and Beyond

DATE	TIME	VENUE
Friday, October 30, 2026	08:30 – 17:10	Texas Governor's Room 3.116, Level 3

Texas Union Building · Austin, Texas · In conjunction with ACM MobiCom 2026

AT A GLANCE

08:30–08:45	Welcome and Opening Remarks
08:45–09:30	Poster Presentations and Q&A
09:30–10:00	Coffee and Networking
10:00–11:00	S1: Drone Security and Attack Detection
11:00–12:00	S2: Trust, Authentication and Resilient Drone Control
12:00–13:30	Lunch and Networking
13:30–14:30	Keynote · Joerg Widmer
14:30–15:30	S3: UAV-Enabled Edge Computing and Resource Allocation
15:30–16:00	Coffee and Networking
16:00–17:00	S4: Learning-Driven Aerial and Non-Terrestrial Networks
17:00–17:10	Closing Remarks

4 technical sessions · 12 full papers · 3 timed poster talks · 1 featured keynote

MORNING PROGRAM · 08:30–11:00

Posters & Drone Security

08:30–08:45 Welcome and Opening Remarks · Workshop Organizers

08:45–09:30 Poster Presentations & Q&A

Three 15-minute poster presentations, including Q&A

POSTER 01

CMA-FL: Cognitive Multi-Agent Federated Learning for Resource-Aware Drone Communication Attack Detection

Pronaya Bhattacharya (Amity University); Somnath Bera (Maulana Abul Kalam Azad University of Technology, West Bengal); Anwesha Mukherjee (Mahishadal Raj College); Debashis De (Maulana Abul Kalam Azad University of Technology, West Bengal); Kwok Tai Chui (Hong Kong Metropolitan University); Brij B. Gupta (Asia University); Thippa Reddy Gadekallu (Zhejiang A&F University)

POSTER 02

Secrecy Rate Analysis of RIS-Assisted UAV Networks Using DDQN Network

Laxmi, Vipul Dixit, Ishan Budhiraja (Bennett University); Swati Kumari (Deakin University); Neeraj Kumar (Thapar University Patiala)

POSTER 03

Distributed Deep Learning-Based CQI Prediction for UAV-Assisted URLLC V2X Networks

Muhammed Mufeed M, Sneha S, Suresh Chavhan (Indian Institute of Science Education and Research, Thiruvananthapuram)

09:30–10:00 Coffee and Networking

10:00–11:00 Session 1: Drone Security and Attack Detection

Full paper presentations and Q&A · 20 minutes each

PAPER 01

Attention-Guided UAV Intrusion Detection with BSO-Based Feature Selection

Akshat Gaurav (Asia University); Varsha Arya (Hong Kong Metropolitan University); Ching-Hsien Hsu (Asia University); Kwok Tai Chui (Hong Kong Metropolitan University); Brij B. Gupta (Asia University)

PAPER 02

Measuring the Safety of Drone Command-and-Control under Link Degradation

Hannaneh B. Pasandi (UC Berkeley); Mohammad Sepahi (Old Dominion University, Rivian Inc.)

PAPER 03

GNSS-Spoofing-Resilient Drone-Assisted 5G Backhaul via Cooperative Detection and Adaptive Link Reconfiguration

Abima Obim Abima, Ali Kashif Bashir, Sara Holm, Muhammad Atif Ur Rehman (Manchester Metropolitan University); MD. Israfil Biswas (University of Bedfordshire); Alex Akinbi (Manchester Metropolitan University)

MIDDAY PROGRAM · 11:00–15:30

Trust & Edge Computing

11:00–12:00 Session 2: Trust, Authentication and Resilient Drone Control

Full paper presentations and Q&A · 20 minutes each

PAPER 01

PDT-TCN: Prefix-Drop Temporal Convolutional Networks for Early Attack Detection in UAV-GCS Communications

Jakub Siłka, Michał Wieczorek, Marcin Woźniak (Silesian University of Technology)

PAPER 02

Post-Quantum Safe Isogeny-Based Authenticated Key Establishment Protocol for Drones-Assisted 6G-Enabled Smart City Environments

Debnath Ghosh (International Institute of Information Technology, Hyderabad); Basudeb Bera (National University of Singapore); Ashok Kumar Das (International Institute of Information Technology, Hyderabad); Biplab Sikdar (National University of Singapore)

PAPER 03

Multi-Factor Trust and Access Control in Drone-Assisted Edge Networks

Deepak Puthal (UAE University); Raja Jayaraman (New Mexico State University); Venkata Prasanth Yanambaka (Texas Woman's University); Pradip Kumar Sharma (University of Aberdeen)

12:00–13:30 Lunch and Networking · Lunch boxes at Cactus Cafe, Level 2

14:30–15:30 Session 3: UAV-Enabled Edge Computing and Resource Allocation

Full paper presentations and Q&A · 20 minutes each

PAPER 01

RAVEN: Reverse Auction-Based Resource Allocation for UAV-Enabled Edge Computing

Ayan Mondal, Mir Razee M., Vansh Kathnawal, Karri Trived (Indian Institute of Technology Indore)

PAPER 02

UAV-Assisted Resource Optimization for Priority-Aware and Trustworthy Services

Muhammad Omair Butt (Lakehead University); Muhammad Naeem (COMSATS University); Waleed Ejaz (Lakehead University)

PAPER 03

Stability-Oriented Multi-Objective Container Scheduling for UAV-Assisted Edge Networks via TD3

Shabir Ahmad (Gachon University); Fahim Ullah Khan, Ibrar Ali Shah (UET Mardan); Abdul Aziz Barnawi, Muhammad Shahid Anwar (King Fahd University of Petroleum and Minerals)

AFTERNOON PROGRAM · 13:30–17:10

Aerial Networks & Closing

13:30–14:30 KEYNOTE ADDRESS

Joerg Widmer · IMDEA Networks Institute, Madrid, Spain

Wireless Sensing for Aerial Networks: From Localization to Coherent Imaging

15:30–16:00 Coffee and Networking

16:00–17:00 Session 4: Learning-Driven Aerial and Non-Terrestrial Networks

Full paper presentations and Q&A · 20 minutes each

PAPER 01

Quantum Deep Reinforcement Learning On-the-Fly: An Energy Efficient Scheme for Autonomous Aerial Vehicles

Anushka Nehra (Thapar University Patiala); Neeraj Joshi (Bennett University); Rajat Chaudhary (Marwadi University); Isaac Woungang (Toronto Metropolitan University)

PAPER 02

Joint Trajectory and Resource Optimization for Dual Aerial ARIS-assisted NOMA-ITNTN

Vangara Saiprudhvi (Vellore Institute of Technology); Keshav Singh (National Sun Yat-sen University); Hariharan Subramaniyam (Vellore Institute of Technology); Chih-Peng Li (National Sun Yat-sen University)

PAPER 03

Adaptive Multi-Band Hybrid RF/FSO Backhaul Optimization for Non-Terrestrial Networks Using Deep Reinforcement Learning

Muhammad Abubakar Rashid, Muhammad Hannan Akram, Wassi Haider Kabir (National University of Sciences and Technology); Haejoon Jung (Kyung Hee University); Syed Ali Hassan (National University of Sciences and Technology)

17:00–17:10 Closing Remarks · Workshop Organizers

The full keynote abstract and speaker biography are provided on the next page.

FEATURED KEYNOTE · 13:30–14:30

Joerg Widmer

Research Professor and Research Director · IMDEA Networks Institute, Madrid, Spain



Joerg Widmer
IMDEA Networks Institute
Madrid, Spain

KEYNOTE ADDRESS

Wireless Sensing for Aerial Networks: From Localization to Coherent Imaging

Wireless sensing, precise localization, and coherent imaging for aerial systems.

TALK ABSTRACT

Next generation wireless communication systems are increasingly capable of providing accurate sensing in addition to connectivity. By reusing spectrum and infrastructure, they can provide capabilities traditionally associated with radar, including precise localization and motion tracking, without dedicated sensing hardware. This is particularly attractive for aerial networks, where payload, energy and spectrum are scarce. Drones play two roles, as platforms that carry wireless nodes and as targets that networks must detect and track. This keynote discusses the evolution of wireless sensing from device localization to coherent imaging in distributed multi-static wireless systems, and what it offers aerial platforms. The talk covers device-free detection and tracking with communication signals, how the time and frequency offsets of unsynchronized devices can be canceled rather than estimated, how sparse frequency bands can be combined to increase resolution, and how many distributed links can cooperate to form synthetic apertures for coherent imaging of dynamic scenes. Because payload power is the limiting resource on a flying platform, it also shows how spike-based receivers reduce the power of the sensing chain several-fold. Throughout, we emphasize experimental testbeds and real-world demonstrations, and highlight the practical challenges and opportunities in turning communication networks into pervasive terrestrial and airborne sensing infrastructures.

SPEAKER BIOGRAPHY

Joerg Widmer is Research Professor and Research Director of IMDEA Networks in Madrid, Spain, where he leads the Wireless Networking Research Group. Previously, he worked at DOCOMO Euro-Labs in Munich, Germany and EPFL, Switzerland, and has held visiting researcher positions at ICSI Berkeley (USA), University College London (UK), and TU Darmstadt (Germany). His research focuses on wireless networks, ranging from extremely high frequency millimeter-wave communication and wireless sensing to mobile network architectures. Joerg Widmer authored over 250 conference and journal papers, and holds 14 patents. He was awarded an ERC consolidator grant, the Friedrich Wilhelm Bessel Research Award of the Alexander von Humboldt Foundation, a Mercator Fellowship of the German Research Foundation, a Spanish Ramon y Cajal grant, as well as 17 best paper awards. He is an IEEE Fellow and Distinguished Member of the ACM.